Response to the Reviewers

Journal of Computer Applications in Archaeology, submission #280, second revision
An Evaluation Artefact in Presence–Background Archaeological Modelling: Evidence from East Java and a Corrected Comparison Protocol

Dear Dr González-Pérez, dear Reviewers,

Thank you for the second round of comments. We have revised the text so that it no longer refers to the review process or to an earlier version of the paper; we have explained the statistical terms for readers who are not statisticians; we have added a new analysis to answer Reviewer 2's question about elevation and volcanic setting; and we have redrawn Figures 1 and 5 and corrected the reference list.

While checking the data for this revision we also found a problem that neither reviewer raised: about a third of the 378 presence records used throughout the paper are modern monuments or misplaced records. We re-ran the real-data analyses on a cleaned subset, and the result changed part of the paper's central claim. We describe this first, under *Changes not requested by the reviewers*, because it is the largest change in this version and the reviewers should see it before the point-by-point answers.

Section, table and figure numbers refer to the revised manuscript.

Changes not requested by the reviewers

1. The presence records, a cleaned subset, and a restated central claim

What we found. The 378 presence records come from OpenStreetMap (279), lists in the Indonesian and English Wikipedia (85) and Wikidata (14). They sit on 335 distinct coordinates. About a third are not archaeological sites: 124 are modern monuments, statues or memorials, and 13 are misplaced or are not sites at all (for example, Central Javanese temples carrying East Javanese coordinates). The previous version presented them, without qualification, as a subset of East Java's archaeological record.

What we did. We defined a cleaning rule and re-ran the analyses on what it keeps. The rule removes 217 of the 378 records: the 85 rows whose coordinates were geocoded from their names (all rows taken from the Wikipedia lists), the 124 modern monuments, and 8 temples of other regions, tourist objects and modern institutions; keeping one row per coordinate removes nothing further. 161 sites remain (155 from OpenStreetMap, 6 from Wikidata). The design of this re-analysis (E236) was committed to the public repository before it was run. It repeated the real-data contrasts the paper's argument rests on (the common-background benchmark, the own- versus common-background matrix, the hard-negative sweep, the priority maps and their seed agreement, and the elevation control), with the analysis code unchanged except for the file of presences it reads.

What changed in the results.

- On the 378 compiled records, redesigning the background with a common evaluation background held fixed contributes −0.014 AUC (interval −0.021 to −0.005 over five seeds), as before.
- On the 161 cleaned sites, the same redesign contributes +0.051 AUC. The pre-registered run used five seeds (+0.0507, positive in 14 of 15 seed×algorithm cells). After that run we extended it to ten seeds, to meet our own rule of at least seven seeds for any reported value, and we report both: +0.051 (+0.039 to +0.063), positive in 29 of 30 cells, and two halves of five seeds that share no seed give +0.0505 and +0.0516. The interval is over seeds, not

over sites; with 161 sites the detection floor is larger than for the 378 records, which the Limitations state.

- On the cleaned sites the hybrid design also beats the random design on every evaluation background (3/3 algorithms on each of the four backgrounds), whereas on the compiled records it led only on its own background.
- What holds in both sets is the inflation from scoring a design on its own background: +0.037 on the compiled records (60/60 cells) and +0.041 on the cleaned sites (54/60). On the cleaned sites, own-background evaluation overstates the redesign gain by +0.047, close to the size of the gain itself.
- The selection criterion still has no interior optimum on the cleaned sites, and the target-group steps of the original sequence are not reproduced in either set. The elevation control gives the same primary result on both sets (below).

How the claim was restated. The previous version concluded that the rise of the original sequence was an evaluation artefact and that, on a common evaluation background, redesigning the background adds nothing (−0.014 AUC). That statement now holds only for the compiled records. The paper now reports both sets and claims what holds in both: own-background evaluation overstates the gain of a background design by about +0.04 to +0.05 AUC, and only a common evaluation background shows how much real gain remains. The original sequence of experiments (E007–E013) is kept on the 378 records, because it is the object under examination.

Where: Abstract; Section 1.3; Section 2.1 (the records and the cleaning rule); new Section 3.3 and Supplementary Table S9 (both sets side by side); Sections 3.2 and 4.1; Conclusions; two new items and one amended item in the Limitations (Section 4.6); Figure 1 (cleaned sites and other records drawn differently); Data Availability (sources, licences and the cleaning script). The abstract therefore differs from the replacement abstract we sent to the editor on 2 October.

2. Other corrections we made ourselves

- **Seed stability.** The previous version measured map agreement against a ten-seed ensemble that contains the seeds being averaged, which flatters the result. We now also report the agreement between ensembles that share no seed (Jaccard 0.76–0.86 for five seeds each on the compiled records, 0.70–0.77 on the cleaned sites) and present "at least seven seeds" as a floor, not as a guarantee of stability (Section 3.8, Figure 6, Table 1).
- **The target-group steps.** The previous version called the whole part of the rise attributed to background design an evaluation artefact. Only the hybrid step is shown to be one; the earlier target-group steps are not reproduced in our benchmark. The text now says so, and the heading of Section 3.2 now reads "The ladder does not survive a common evaluation background" (Sections 1.3 and 3.2).
- **"Pre-registered".** Of the experiments E217–E224, only E224 had its design committed before the run; E218 and E220–E223 were run from written designs that entered the repository together with their results, and E217 and E219 had no written design. Section 2.6 states this, and "pre-registered" is now used only for E224 and E235 (Sections 2.6 and 3.5).
- **Synthetic inflation by design.** The count "343/360" pooled the three designs. It is now given per design as well: hybrid 240/240 runs, random 52/60, target-group 51/60 (Section 3.2, Figure 3).
- **Robust core.** Its definition was stated wrongly; it is the set of cells in the top decile under all three background designs, and the caveat now gives the right reason why the density

contrast is not a validation (the same sites trained the models) (Section 3.8).

- **Share of hard negatives.** The realised share of hard negatives in the hybrid design (0.62, against a target of 0.30) was attributed to the target-group pool. The proportion of such points in the target-group background (0.485) is close to that of the random background (0.503), so the explanation is now the composition of the whole background pool (Section 2.3).
- **Test 3.** The previous version used a performance check across road-distance quartiles ("Test 3"). It was scored on each design's own background, so it is subject to the artefact this paper describes; the text now says so and does not use it (Section 2.4).
- **Data description.** The Data Availability statement listed a source that contributed no records and gave a wrong licence for Wikidata; both are corrected.
- **Authors' contributions and Declaration of AI use.** The Authors' Contributions now say that the analyses were implemented with AI assistance, and the Declaration of AI Use states how the analyses and the text were produced, in the terms of our message to the editor of 2 November.

Reviewer 1

R1-1

Comment (summarised). The paper should be self-contained: no mention of reviewers, requests, responses, revisions or an earlier version; the story of the rework could go in the acknowledgements.

Response. We removed or rewrote every passage of the main text that referred to the reviewers, to the review process or to an earlier version of the paper (38 in the previous version), including the abstract. The development of the analysis is now presented as the study's own sequence of experiments (Sections 1.3, 2.3 and 3.1), and the later experiments as tests of that sequence. Outside the main text, the Acknowledgements keep the conventional thanks to the editor and reviewers, and the Declaration of AI Use describes how the earlier drafts were produced, as the journal's AI policy requires.

Where: whole text; Abstract; Acknowledgements; Declaration of AI Use.

R1-2

Comment (summarised). The research question should be stated earlier, more clearly and more pedagogically for readers who are not statisticians.

Response. The first paragraph of the Introduction now states the question in plain terms: a change in how background points are selected can raise the cross-validated AUC without the model becoming better, because the change can make the evaluation task easier; the study tests which of the two is happening.

Where: Section 1, first paragraph.

R1-3

Comment (summarised). The statistical methods need more explanation for non-specialists.

Response. We added Table 1, *Key terms*, which defines in plain language the terms the paper relies on: presence–background model, background design, hard negatives, evaluation background,

availability domain, AUC, TSS, continuous Boyce index, spatial block cross-validation, seed ensemble, Jaccard overlap and coarsened exact matching.

Where: Table 1 (Section 1.1).

R1-4

Comment (summarised). Three expressions are hard to understand.

Response. All three are replaced by literal wording: "we sweep the hard-negative dial" now reads "we vary the share of hard negatives in the background from 0 to 1" (Section 2.6); "how large is judging at home?" now reads "how large is own-background inflation?" (Section 3.2); and "map Jaccard said no" now reads "the pre-registered primary metric was map Jaccard, and it did not support that reading" (Section 3.6). We also removed the rest of the same family of expressions throughout the text and in the figures (for example "dial", "home-court" and "judged at home"). The word "ladder" is kept for the original sequence of seven experiments and is defined where it first appears (Section 1.3).

Where: Sections 1.3, 2.6, 3.2 and 3.6; Figures 2–4.

R1-5

Comment (summarised). The section sign is unusual.

Response. All section references now read "Section n".

Where: whole text.

R1-6

Comment (summarised). The bibliography is inconsistently formatted and some names are misspelt; the Comer et al. reference is wrong.

Response. We checked every reference that has a DOI against Crossref. Three entries carried wrong metadata and are corrected: Comer et al. (2023) (authors, title and volume), Wang, Shi & Oguchi (2023) (authors and title) and Castiello & Tonini (2021) (title and the first author's given name). The sentence that cited these three papers together did not describe them accurately; it has been rewritten from what the papers report (Section 1.3). Name spellings were corrected, DOIs were added where they exist, title capitalisation was made consistent, and the whole list now follows the journal's Harvard style.

Where: References; Section 1.3.

R1-7

Comment (summarised). Figure 1 should show the study area; it is a scatter plot whose axis values cannot be read.

Response. Figure 1 is now a relief map drawn from the elevation model used by the models, with the coastline, a latitude–longitude grid, a scale bar, a north arrow and a locator map of Indonesia. It shows the presence records (the 161 cleaned sites and the other 217 records drawn differently), the 13 volcanic centres inside the study area, and the 20 km and 40 km distance lines used in the new elevation control.

Where: Figure 1.

R1-8

Comment (summarised). The legend of Figure 5 should not sit on the map.

Response. The legend is now below the map. The map is drawn as a raster with the same coastline, grid and scale bar as Figure 1.

Where: Figure 5.

R1-9

Comment (summarised). The text should be cleaned of the history of the previous version.

Response. As for R1-1.

Reviewer 2

R2-1

Comment (summarised). It is still unclear whether the model responds to volcanic areas or simply to high elevation; comparing volcanic land with non-volcanic land at the same elevation would settle it.

Response. We ran the comparison you suggest as a new analysis (E235), with its design committed to the public repository before it was run. Land is "near" if it lies within 20 km of a Quaternary volcanic centre and "far" if it lies 40 km or more from one; within each 200 m elevation band the two are matched on slope, terrain ruggedness and topographic wetness, and uncertainty comes from a spatial block bootstrap (1,000 replicates). The result has two parts.

- **Above about 1,000 m the comparison cannot be made in East Java.** Of the roughly 3,435 km^2 of land above 1,000 m, 97.5% lies within 20 km of a Quaternary volcanic centre and only 0.4% lies 40 km or more from one (about 15 km^2 at 1,000–1,200 m, none above 1,200 m). Elevation and volcanic setting therefore cannot be separated there, and the paper now says so and attributes the pattern above 1,000 m to neither.
- **Below 1,000 m no direction is claimed.** With XGBoost, the primary learner, matched suitability is +0.049 higher near volcanic centres for the random design, +0.049 for the target-group design and +0.060 for the hybrid design. Under the pre-registered rule the hybrid result alone meets the criterion for a volcano-specific difference, and the summary across the three designs is inconclusive. For the random design the two pre-registered sensitivity definitions give −0.002 and +0.016, and post hoc variants give +0.005 to +0.07. The far uplands lie mostly in the Southern Mountains, an older (Eocene–Miocene) volcanic arc, so they are not strictly non-volcanic land, and the text states this.

No model input carries volcanic information. We removed the earlier terrain-matched comparison, which had only two sites in its non-volcanic arm, and the wording that its result was consistent with a volcano-specific effect; the new analysis replaces it, and the paper no longer implies that the model's response is volcano-specific. On the cleaned sites the primary result is the same (the hybrid design alone meets the criterion; the summary is inconclusive), while the two sensitivity definitions exchange their labels between inconclusive and not specific (Section 3.3).

Where: new Section 3.10; Figure 7; Supplementary Tables S7–S8; Section 4.6 (Limitations); Section 3.3.

R2-2

Comment (summarised). Without topography in Figure 1, readers cannot see how the model relates to elevation.

Response. The new Figure 1 shows elevation (a hillshaded elevation model with an elevation scale) together with the distance lines used in the new control.

Where: Figure 1.

We thank both reviewers again. The remaining errors are ours.

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